

Experiment 1

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Branch: BE-CSE

Section/Group: 801-B

Semester: 6th

Date of Performance:13/01/25

Subject Name: System design

Subject Code: 23CSH-314

1. Aim:

To design, implement, and evaluate a simple URL redirection system that converts long URLs into short URLs, and to measure system performance (latency and throughput) under varying request loads.

2. Objective:

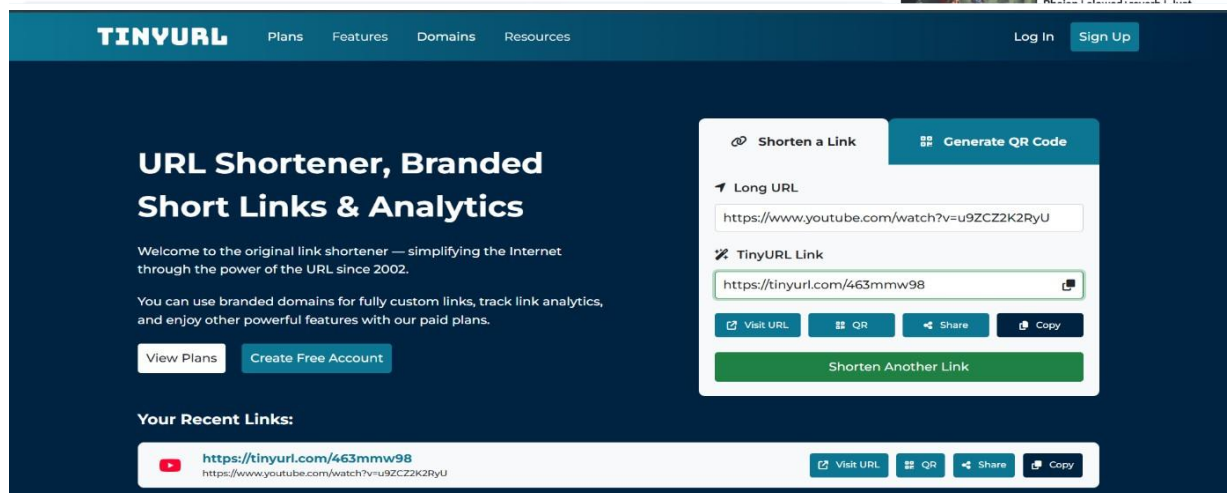
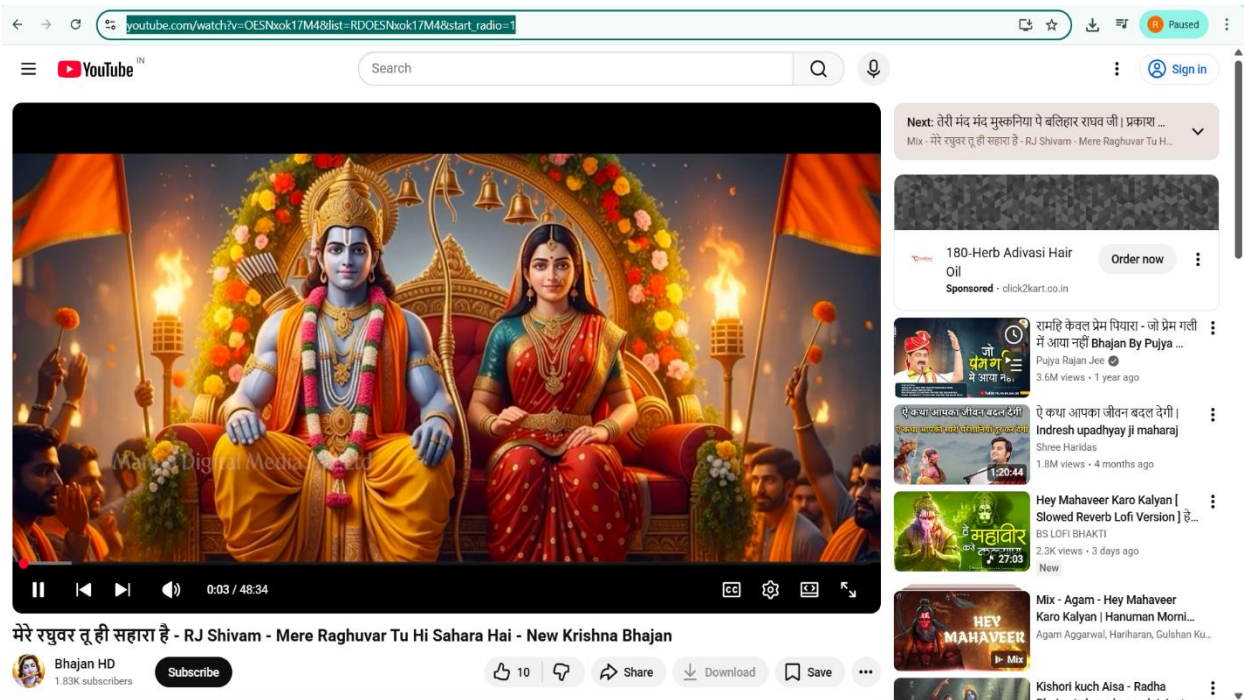
- To understand the working principle of a URL shortening and redirection system.
- To study how long URLs are converted into short, unique identifiers.
- To analyse redirection behaviour and response time of a URL shortening service.
- To evaluate system performance using metrics such as latency and throughput under different request loads.
- To gain practical exposure to real-world system design using an existing URL shortening tool.

3. Tools Used: TinyURL

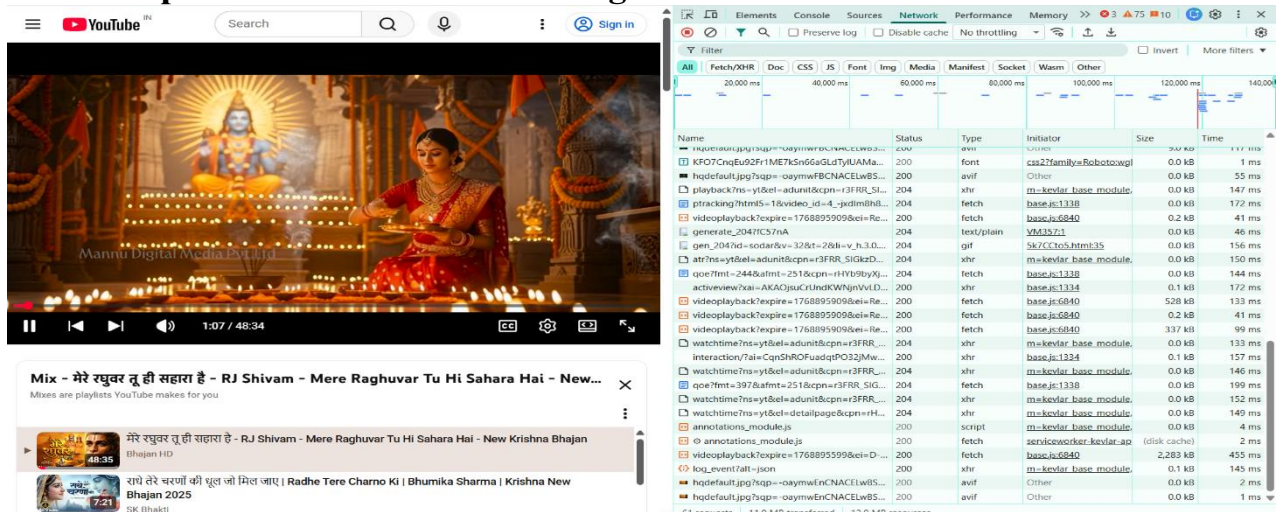
4. Procedure:

Step 1: URL Shortening

- Open the TinyURL website.
- Copy a long URL (e.g., an academic website or documentation page).
- Paste the long URL into the TinyURL input field.
- Click “Make TinyURL!”
- Note the generated short URL.



Step 2: URL Redirection Testing



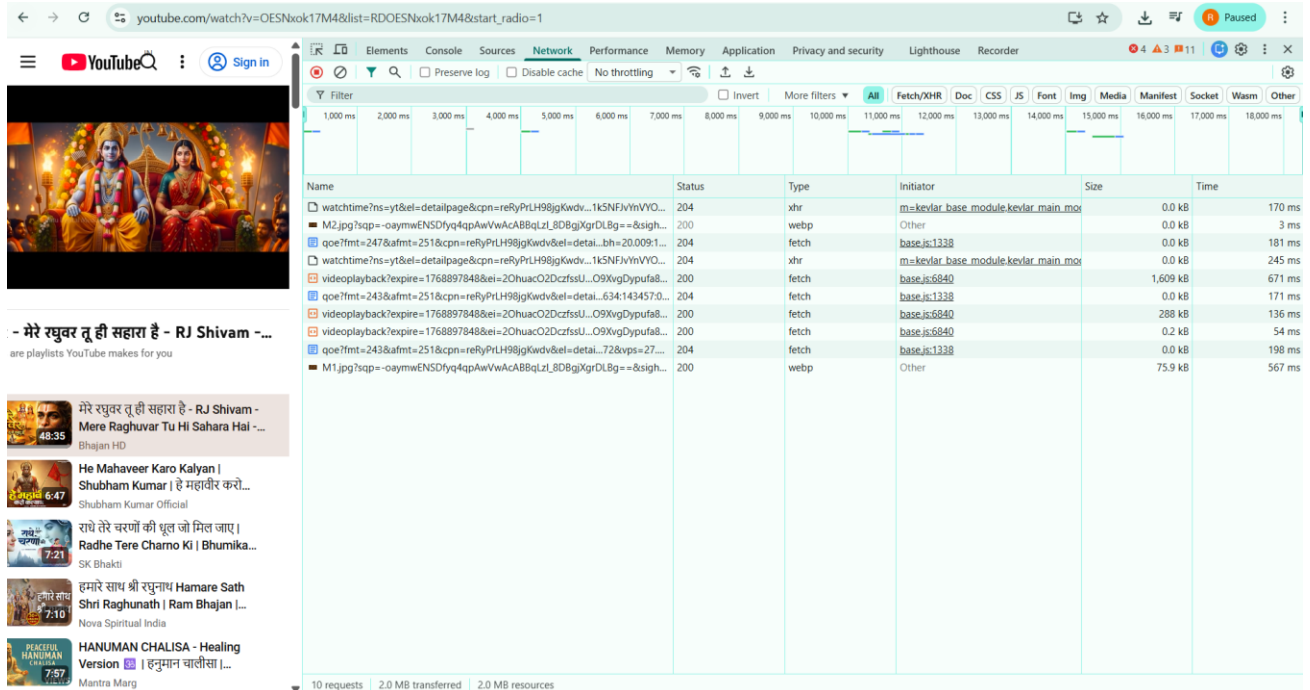
- Paste the generated short URL into a new browser tab.
- Observe the redirection to the original long URL.
- Confirm correctness of redirection.

Step 3: Latency Measurement

- Record the time just before pressing Enter on the short URL.
- Record the time when the destination page fully loads.
- Calculate latency.
- Repeat this process 5–10 times.
- Compute Average Latency.

Step 4: Throughput Measurement

- Send multiple requests to the same short URL within a fixed time interval (e.g., 60 seconds).
- Count the total number of successful redirections.
- Repeat with increased request loads (manual refresh or multiple tabs).



youtube.com/watch?v=OESNxok17M4&list=RDOESNxok17M4&start_radio=1

YouTube Sign in

Local metrics

- Largest Contentful Paint (LCP)**
1.72 s
Your local LCP value of 1.72 s is good.
LCP element [video.video-stream.html5-main-video](#)
- Cumulative Layout Shift (CLS)**
0.05
Your local CLS value of 0.05 is good.
Worst cluster [9 shifts](#)
- Interaction to Next Paint (INP)**
16 ms
Your local INP value of 16 ms is good.
INP interaction [pointer](#)

[Learn more about local and field metrics](#)

Interactions | Layout shifts

Interaction	Time
pointer INP	16 ms
pointer video.video-stream.html5-main-video	16 ms

Next steps

Field metrics
See how your local metrics compare to real user data in the [Chrome UX Report](#).
[Set up](#)

Environment settings
Use the [device toolbar](#) and configure throttling to simulate real user environments and identify more performance issues.

CPU: No throttling | Network: No throttling | ☐ Disable network cache

[Record](#) | Ctrl + E

[Record and reload](#) | Ctrl + Shift + E

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Network

Filter | 5,000 ms | 10,000 ms | 15,000 ms | 20,000 ms | 25,000 ms | 30,000 ms | 35,000 ms | 40,000 ms | 45,000 ms

Name	Status	Type	Initiator	Size	Time
watchtime?ns=yt&el=detailpage&cpn=reflyPrLH98jgKwDv...1k5NFjvYmVYO...	204	xhr	m=kevar_base.module.kevar_main.mo	0.0 kB	170 ms
M2.jpg?sqp=--oaymwENSDFyq4pAwVwAcABqLzL_BDBgKgrDLBg=--&sig=...	200	webp	Other	0.0 kB	3 ms
qoe?fmt=247&afmt=251&cpn=reflyPrLH98jgKwDv&el=detail...bh=20.0091...	204	fetch	base.js:1338	0.0 kB	181 ms
watchtime?ns=yt&el=detailpage&cpn=reflyPrLH98jgKwDv...1k5NFjvYmVYO...	204	xhr	m=kevar_base.module.kevar_main.mo	0.0 kB	245 ms
videoplayback?expire=1768897848&ei=2OhuacO2DczfsU...09XvgDypufa8...	200	fetch	base.js:5840	1,609 kB	671 ms
qoe?fmt=243&afmt=251&cpn=reflyPrLH98jgKwDv&el=detail...634:1434570...	204	fetch	base.js:1338	0.0 kB	171 ms
videoplayback?expire=1768897848&ei=2OhuacO2DczfsU...09XvgDypufa8...	200	fetch	base.js:5840	288 kB	136 ms
videoplayback?expire=1768897848&ei=2OhuacO2DczfsU...09XvgDypufa8...	200	fetch	base.js:5840	0.2 kB	54 ms
qoe?fmt=243&afmt=251&cpn=reflyPrLH98jgKwDv&el=detail...72&vps=27...	204	fetch	base.js:1338	0.0 kB	198 ms
M1.jpg?sqp=--oaymwENSDFyq4pAwVwAcABqLzL_BDBgKgrDLBg=--&sig=...	200	webp	Other	75.9 kB	567 ms
watchtime?ns=yt&el=detailpage&cpn=reflyPrLH98jgKwDv...1k5NFjvYmVYO...	204	xhr	m=kevar_base.module.kevar_main.mo	0.0 kB	164 ms
watchtime?ns=yt&el=detailpage&cpn=reflyPrLH98jgKwDv...1k5NFjvYmVYO...	204	xhr	m=kevar_base.module.kevar_main.mo	0.0 kB	158 ms
videoplayback?expire=1768897848&ei=2OhuacO2DczfsU...09XvgDypufa8...	200	fetch	base.js:5840	1,447 kB	707 ms
videoplayback?expire=1768897848&ei=2OhuacO2DczfsU...09XvgDypufa8...	200	fetch	base.js:5840	1,100 kB	307 ms
qoe?fmt=243&afmt=251&cpn=reflyPrLH98jgKwDv&el=detail...09:2547311:1...	204	fetch	base.js:1338	0.0 kB	168 ms

15 requests | 4.5 MB transferred | 4.6 MB resources

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[Record and reload](#) | Ctrl + Shift + E

5. Calculations:

Latency: Page load time- Request time

Throughput: no. of request served/ time

Table: Observation Table

Record no.	No. of requests	Data transfer (Mb)	Time window in sec	INP/ Latency(m/s)	Throughput Mb/s
1.	10	2	18	16	0.111
2.	15	4.5	45 s	16	0.1

NOTE: Latency, here is calculated directly from INP. INP directly measures the time between user interaction and visible response.

6. Result:

The URL redirection system demonstrated efficient performance under varying request loads. The measured latency was low, indicating fast redirection, while throughput values confirmed the system's ability to handle multiple concurrent requests without failure. The system remained stable with minimal layout shift and consistent response times.

7. Learning Outcomes:

- Understand the internal working of a URL redirection system.
- Use browser developer tools to evaluate real-world system performance.
- Measure latency using Web Vitals such as INP and LCP.
- Calculate throughput using network request analysis.
- Interpret performance metrics for system scalability and reliability.